Sahaj Bhadja

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Summary

Software engineer with 2+ years building high-performance data pipelines, simulation systems, cross-platform software, and reliable backend services using C++, Python, Linux, algorithms, multithreading, automated testing, CI/CD, and machine learning.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant PostgreSQL, Azure SQL, and Blob Storage services for 500+ users, optimizing schemas and indexed queries to reduce high-volume dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant **FastAPI orchestration** for 1,000+ monthly jobs using explicit states, retries, validation, and failure isolation, tripling throughput while strengthening production reliability and debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Python APIs and concurrent pipelines** processing 10,000+ daily submissions, integrating secure file ingestion and AWS RDS persistence while maintaining stable performance under production workloads.
- Designed multithreaded batch processing for 1,000+ files with fault-tolerant execution, automated validation, and failure recovery, reducing processing turnaround by 60% while improving system observability and testability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Decision Transformer environment for **autonomous-driving simulation**, processing 100,000+ trajectory logs and improving long-horizon planning by 18% through reproducible data pipelines and systematic performance evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and historical trajectories through multi-head attention, improving closed-loop simulation stability by 15% across complex, high-volume physical-world scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform Python microservices** and 20+ PyTest and Mockito suites through Jenkins CI/CD, automating unit and integration validation across Windows, Linux, and macOS production environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular platform with REST APIs, WebSockets, and eight Node.js services, supporting 2M+ synchronized records while translating complex requirements into maintainable real-time software workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent production releases.

Skills

Systems & Languages: C++, Python, C, Java, JavaScript, TypeScript, SQL, Bash, Linux, Unix, data structures, algorithms, object-oriented programming, design patterns, concurrency, multithreading, distributed systems, system design, performance optimization, memory management; **Data & Computing:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, ETL, data acquisition, data pipelines, data validation, geometry processing, simulation, machine learning, deep learning, PyTorch, Transformers, computer vision, numerical analysis; **Engineering:** FastAPI, REST APIs, AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, regression testing, debugging, profiling, monitoring, fault tolerance, Agile/Scrum, technical documentation, cross-functional collaboration

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six **CPU scheduling algorithms** with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ process combinations while reducing algorithm-debugging time by 20%.

PiXMod Live | *Python, PyTorch, SAM, Stable Diffusion, MongoDB*
- Built an AI image-processing system using SAM for segmentation and Stable Diffusion for geometric manipulation, processing 1,000+ assets while maintaining scalable image and metadata storage in MongoDB.

CaptionCraft Live | *Python, PyTorch, Transformers, Flask, Gradio*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom transformer models on 10,000 images and 100,000 captions, improving model accuracy from 81% to 96% through systematic experimentation and evaluation.